1. Component Checklist

This is everything you'll need, broken down into hardware and software.

Hardware Components ⚙️

- **Embedded System (Inside the Bat):**
  - **Microcontroller (MCU): ESP32-WROOM-32**. It's the perfect choice for its power, low cost, and built-in Wi-Fi/Bluetooth.
  - **Motion Sensor (IMU): BNO055** (9-axis) or **MPU-6050** (6-axis). The BNO055 is slightly better as it has an internal fusion algorithm, making swing path calculations easier.
  - **Impact Sensors: 3 to 4 small Piezoelectric Disc Sensors**. These will be placed on the bat face to detect the exact impact point.
  - **Power:** A **400-500mAh LiPo Battery** coupled with a **TP4056 charging module** with a **USB-C port** for modern, convenient charging.
  - **Casing:** A custom-designed, shock-absorbent housing (likely 3D printed or molded) to protect the electronics.
- **External Hardware:**
  - **The Camera:** The user's own **Smartphone** (e.g., iOS 14+ or Android 9+).
  - **Stabilizer:** A basic **Tripod** for the smartphone.

Software & Technology Stack 💻

- **Firmware (For the ESP32):**
  - **Language: C++** using the Arduino framework for rapid development.
- **Mobile Application (For the Smartphone):**
  - **Framework: Flutter** or **React Native**. These are cross-platform, allowing you to build for both Android and iOS from a single codebase, saving significant time and resources.
  - **Communication Protocol: Bluetooth Low Energy (BLE)** for connecting the app to the bat.
- **Artificial Intelligence:**
  - **Computer Vision Library: Google's MediaPipe**. It's highly optimized for mobile devices and provides excellent pre-built models for **Pose Estimation**.
- **Backend & Database (Cloud):**
  - **Platform: Firebase (by Google)**. It provides a simple solution for user authentication (login), a database (Firestore) to store session data, and cloud storage for video clips.

2. Complete Data & System Workflow

This is the step-by-step journey of data from the player's action to the final insight.

1. **Setup:** The player opens the **Coach's Eye App**, places their phone on a tripod at the bowler's end, and connects their Smart Bat via **Bluetooth**. They then start a "Video Analysis Session."
2. **Action (Swing & Impact):** The player swings the bat and hits the ball.
    a. The **IMU** sensor captures the entire 3D motion of the swing (acceleration and rotation data).
    b. The **Piezoelectric sensors** trigger at the exact moment of impact, noting the vibration intensity at different points to determine the hit location (sweet spot or edge).
3. **Data Transmission:** The **ESP32** microcontroller inside the bat instantly processes this raw data into a compact packet (e.g., swing arc data + impact timestamp + location data) and transmits it to the smartphone app via **BLE**. This happens in milliseconds.
4. **Synchronization:** The app receives this data packet. The **impact timestamp** is the crucial piece of information. The app's software immediately matches this timestamp to the corresponding frame in the video it is recording.
5. **AI Analysis (Post-Shot Processing):** Once the shot is complete, the app isolates the relevant video clip (e.g., 1 second before and 1 second after impact). It then runs the **MediaPipe Pose Estimation** model on this clip to analyze the player's technique (head position, footwork, elbow angle).
6. **Insight Generation:** The app's algorithms **fuse** the two data sources:
    a. **Bat Data:** Calculates final metrics like **Bat Speed (km/h)**, **Power Index**, and **Sweet Spot Accuracy (%)**.
    b. Video Data: Generates technique feedback like **"Head was still"** or **"Improve front foot placement."**
7. **Display:** The result is presented to the user as an interactive, slow-motion video replay of their shot, complete with overlaid stats and actionable AI-powered coaching tips. This data is then saved to their profile in the **Firebase** cloud.

3. Project Execution Roadmap

This is a phased plan to take you from idea to a market-ready product.

Phase 1: Proof of Concept (2 Weeks)

- **Goal:** Validate that the core technology works.
- **Tasks:**
    - Acquire developer boards (ESP32, MPU-6050) and sensors.

    - Tape the hardware securely onto a standard cricket bat.
    - Write basic firmware to read sensor data on swing and impact.
    - Build a minimal app that can connect via BLE and simply display the raw sensor data in real-time.

Phase 2: Prototyping & Development (4 Weeks)

- **Goal:** Create the first integrated prototype and the core app.
- **Tasks:**
    - Design a custom PCB to make the electronics compact.
    - Design and 3D print the protective housing.
    - Work with a bat maker to embed the housing into a bat prototype.
    - Develop the main User Interface (UI) of the mobile app and establish a stable BLE connection.
    - Set up the Firebase backend for user accounts.

Phase 3: AI Integration & Algorithm Refinement (2 Weeks)

- **Goal:** Build the "brains" of the system.
- **Tasks:**
    - Integrate the MediaPipe library into the app for pose estimation.
    - Develop the crucial synchronization algorithm to link video frames with bat data.
    - Write the code that converts raw sensor numbers into meaningful cricket metrics (this will require testing and calibration).

Phase 4: Field Testing & Calibration (2 Weeks)

- **Goal:** Ensure the product is accurate, durable, and user-friendly.
- **Tasks:**
    - Give the prototype bats to a group of local cricketers for beta testing.
    - Collect extensive data and user feedback.
    - Use high-speed cameras to calibrate your Bat Speed algorithm for accuracy.
    - Fix bugs in both the firmware and the app.

Phase 5: Launch Readiness (2 Weeks)

- **Goal:** Prepare for production and market entry.
- **Tasks:**
    - Finalize the hardware design for manufacturing.
    - Submit the app to the Google Play Store and Apple App Store.
    - Prepare marketing materials.

- Launch the product.